

Cluster Quilting: Spectral Clustering for Patchwork Learning

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Joint work with Andersen Chang, Genevera I. Allen

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Motivation

Learn from data observed in patches: **subsets of both observations and features** (patches) (Rajendran et al., 2023)

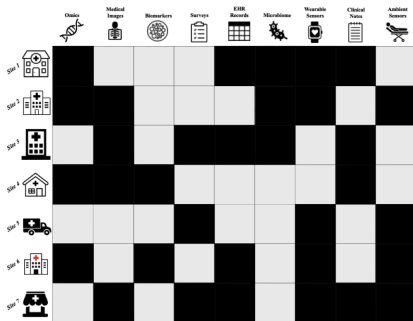
Patchwork Learning

Learn from data observed in patches: **subsets of both observations and features** (patches) (Rajendran et al., 2023)



Patchwork Learning

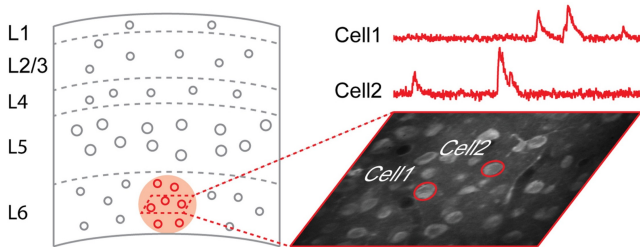
Learn from data observed in patches: **subsets of both observations and features** (patches) (Rajendran et al., 2023)



Patchwork Learning in Healthcare (Rajendran et al., 2023)

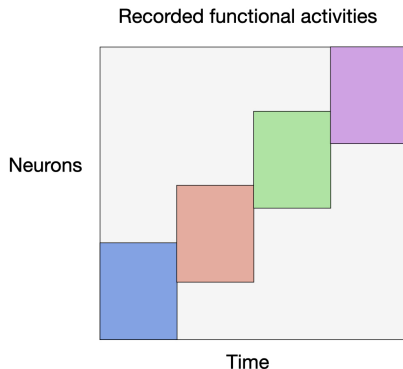
Patchwork Learning

Learn from data observed in patches: **subsets of both observations and features** (patches) (Rajendran et al., 2023)



Calcium imaging in neuroscience

Learn from data observed in patches: **subsets of both observations and features** (patches) (Rajendran et al., 2023)



Learn from data observed in patches: **subsets of both observations and features** (patches) (Rajendran et al., 2023)

Many other applications: data integration, survey data, finance data, etc

Statistical analysis for patchwork data is generally challenging:

- Missingness is not uniform at random; certain feature blocks (modalities) completely missing for observation groups
- Limited studies on matrix completion with deterministic, blockwise missing patterns (Chatterjee, 2020; Yan and Wainwright, 2024).
- Downsampling is the common practice

Clustering for Patchwork Learning

In this talk, focus on the **clustering** task. Given patchwork observations, can we find comprehensive clustering structure for all samples?



Many sample pairs are never observed on the same feature set: significant missingness also occurs in similarity/distance matrix.

Clustering for Patchwork Learning

Prior Works:

- Delete + clustering: fail to give comprehensive clustering of all samples
- Impute + clustering (Aktaş et al., 2019): no statistical guarantee.
- Cast as likelihood-based optimization problems (Chi et al., 2016; Poddar and Jacob, 2019); non-convex, no algorithmic convergence guarantees with such observational pattern

Our goal: develop a novel clustering algorithm for patchwork learning that is **statistically accurate, theoretically sound, and efficient.**

Idea: leverage the idea of [spectral clustering](#)

Cluster Quilting: Model Set-up and Proposed Algorithm

- Sub-Gaussian mixture model:
 - $K > 1$ clusters, centroids $\theta_1^*, \dots, \theta_K^* \in \mathbb{R}^p$
 - n samples: $z_1, \dots, z_n \in [K]$ are cluster labels
 - Data $X_i = \theta_{z_i} + \epsilon_i \in \mathbb{R}^p$, ϵ_i is mean-zero noise
 - Data matrix $X = (X_1, \dots, X_n)^\top \in \mathbb{R}^{n \times p}$ is of low-rank mean (rank $r \leq K$)

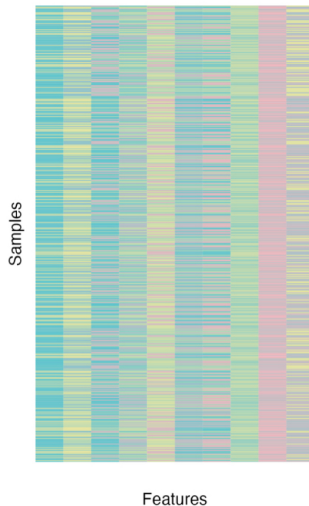
- Patchwork observational pattern
 - We observe M data patches, each indexed by feature subset $T_m \subset [p]$ and sample subset $I_m \subset [n]$: $X_{I_m, T_m} \in \mathbb{R}^{n_m \times p_m}$
 - W.L.O.G., T_m 's are disjoint sets of column indices, $\cup_{m=1}^M T_m = [p]$, and $\cup_{m=1}^M I_m = [n]$.
- Our goal is to estimate the cluster labels $z_1, \dots, z_n$ up to permutation on $[K]$.

Problem Set-up

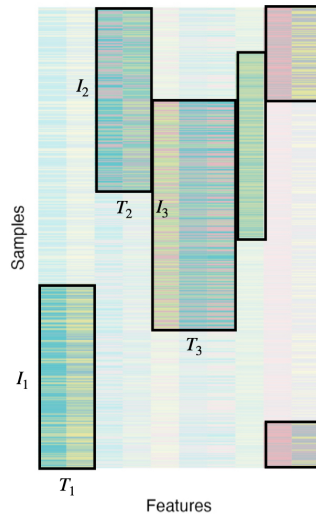
Data organized by clusters



Full data if observable

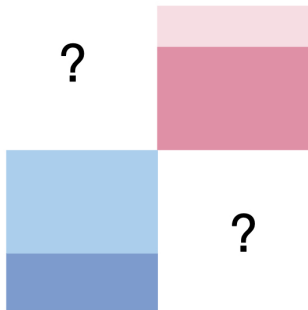


Patchwork observations



Problem Set-up

Is comprehensive clustering possible in patchwork learning? Not always!



Hard example: observed data



One possible complete data

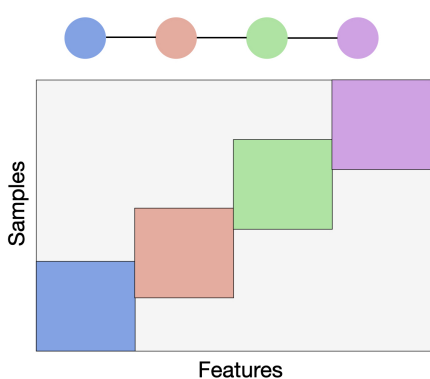


Alternative complete data

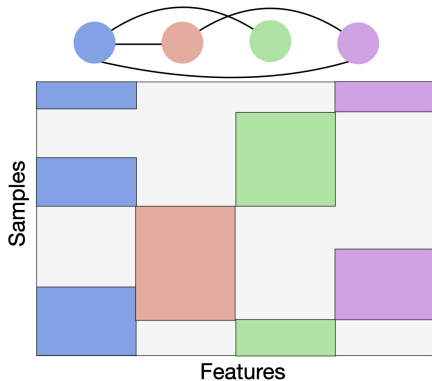
Insight: need information sharing across patches through overlapping observation sets.

Problem Set-up

Graph representation of patchwork observational pattern:



(a) Sequential patches (calcium imaging)



(b) Mosaic patches (healthcare and genomics)

Assume connected graph!

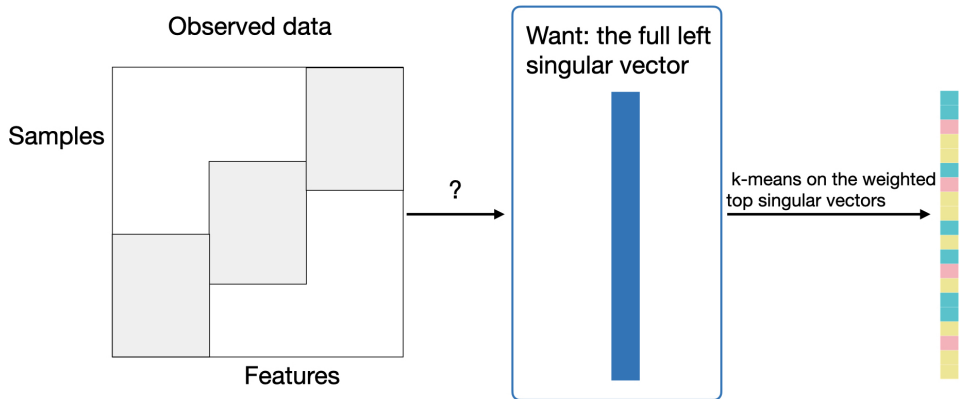
Our Algorithm: Cluster Quilting

Idea:

- Want to estimate the **top singular vectors** of the whole data matrix (spectral clustering)
- SVD for each patch only estimates some rows
- Combine the SVD results from each patch $\Rightarrow$ full top singular vectors

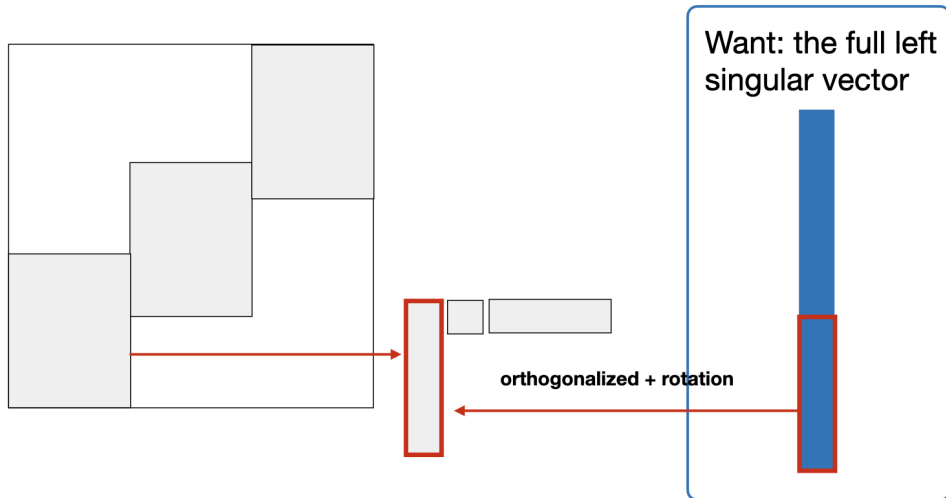
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Start from sequential observational pattern:



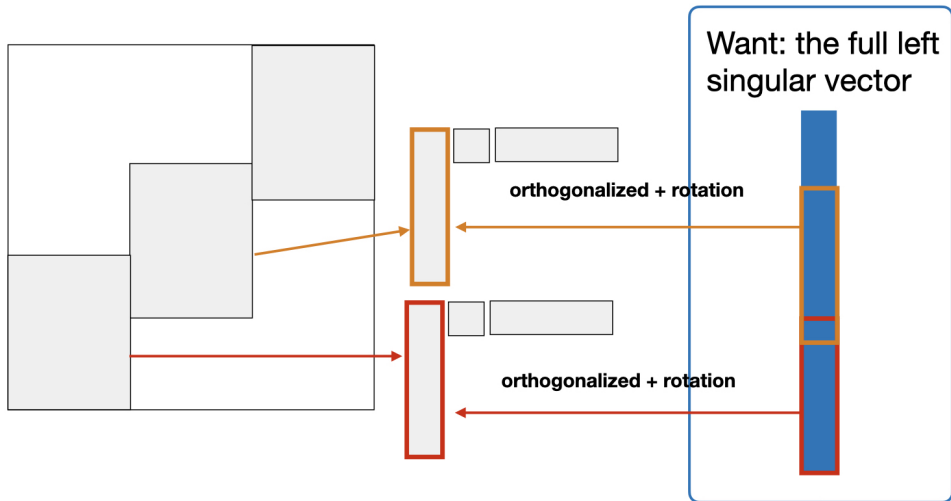
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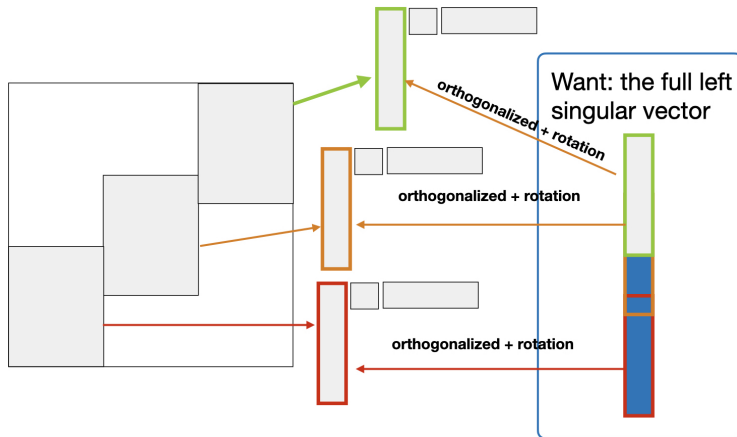
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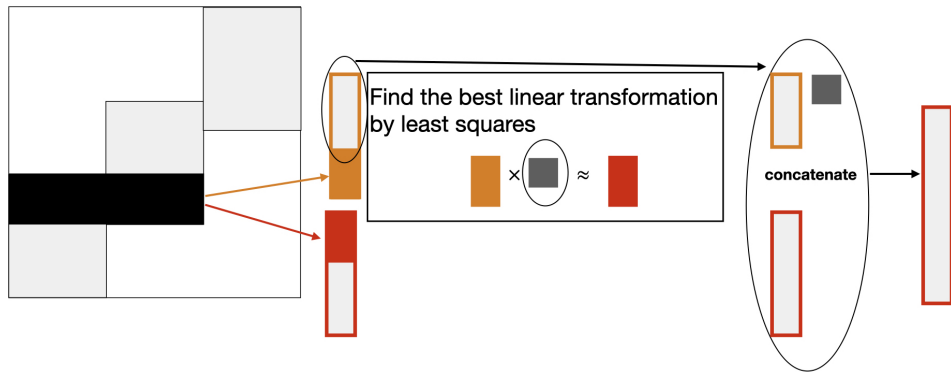
Start from sequential observational pattern:



Need to find linear transformations to match patches!

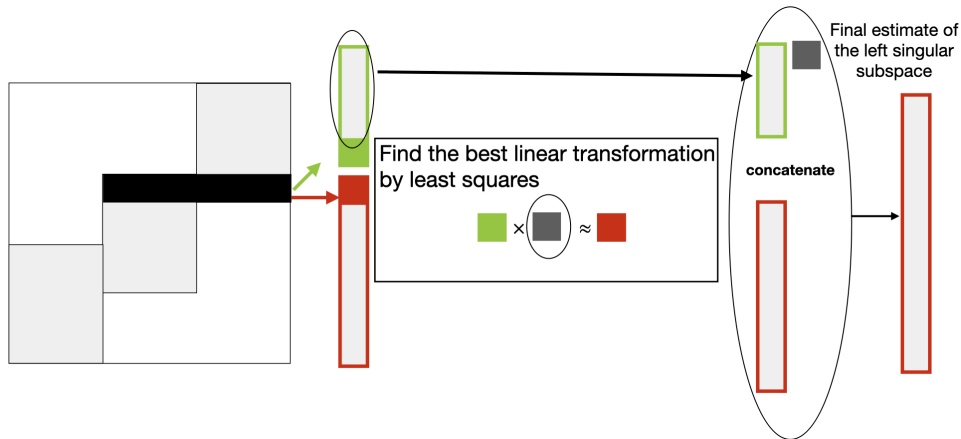
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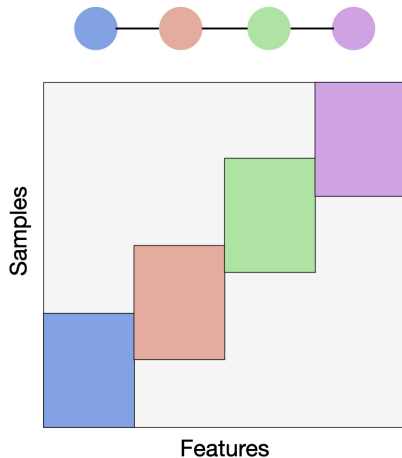
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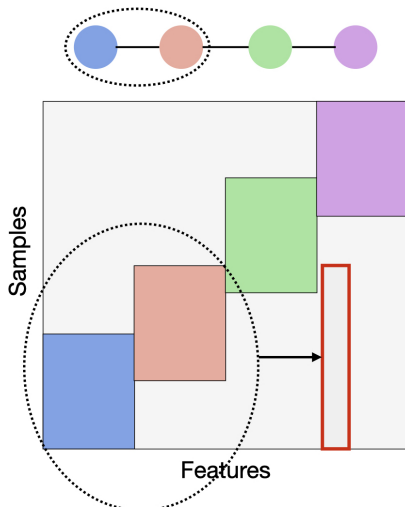
Our Algorithm: Cluster Quilting

What about non-sequential patches? Another look at the procedure:



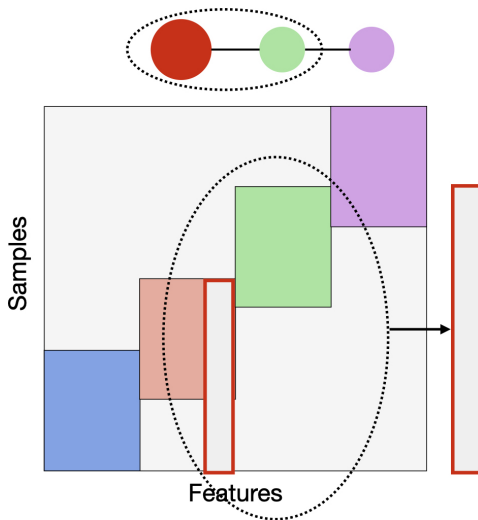
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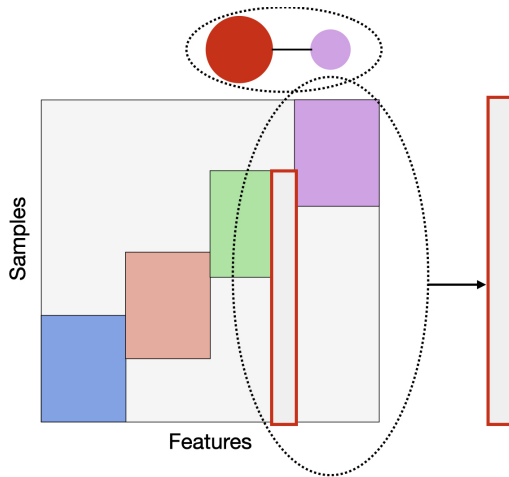
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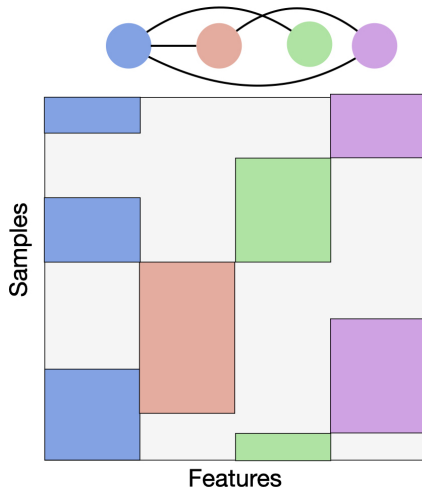
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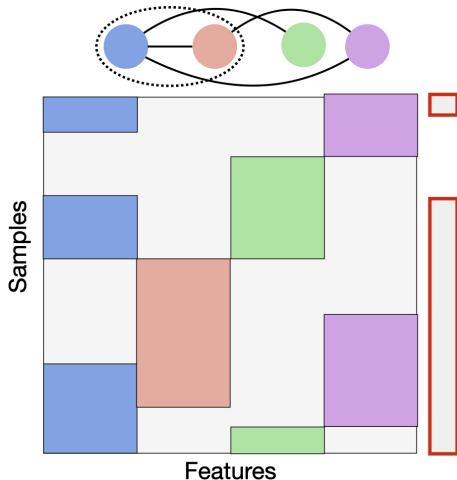
Our Algorithm: Cluster Quilting

Similarly, we can merge nodes in the graph one by one in general patchwork settings:



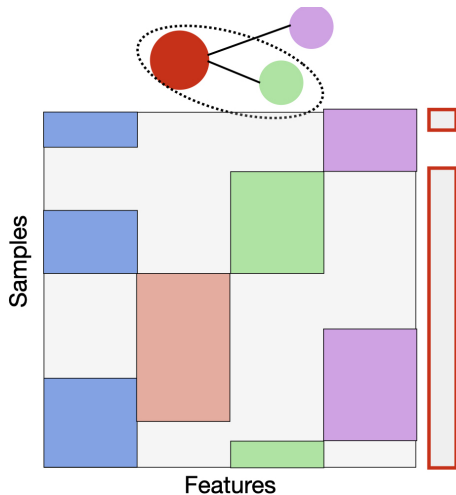
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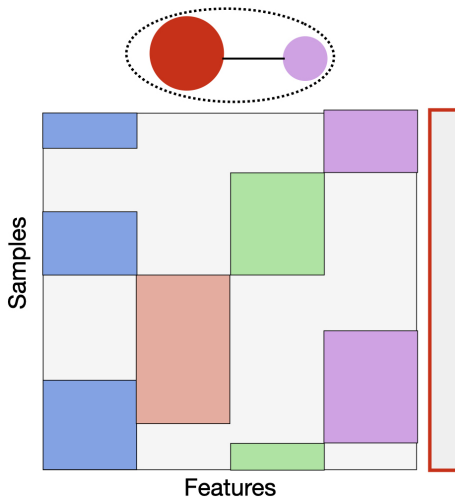
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Our Algorithm: Cluster Quilting

Similarly, we can merge nodes in the graph one by one in general patchwork settings:



Summary:

1. Perform SVD for each data patch
2. Find an ordering of data patches along a path in the patch graph
3. Match the top singular vectors of later patches to the first patch, sequentially

Want sufficient overlap between each patch and its prior patches. Given a patch ordering $\pi : [M] \rightarrow [M]$, let $s(\pi(m), \cup_{l < m} l_{\pi(l)})$ be a measure for the overlapping signal between m th patch and prior patches; $\cup_{l < m} l_{\pi(l)}$ is the sample set covered by prior patches.

Consider the ordering $\hat{\pi}$:

$$\hat{\pi} = \arg \max_{\pi: [M] \rightarrow [M]} \prod_{m=2}^M s(\pi(m), \cup_{l < m} l_{\pi(l)}), \quad (1)$$

$s(\pi(m), \cup_{l < m} l_{\pi(l)})$ will be chosen based on theory.

Want sufficient overlap between each patch and its prior patches.

$$\hat{\pi} = \arg \max_{\pi: [M] \rightarrow [M]} \prod_{m=2}^M s(\pi(m), \cup_{l < m} l_{\pi(l)}), \quad (1)$$

Combinatorial problem; hard for large M .

For speed-up, greedy search:

- $(\hat{\pi}(1), \hat{\pi}(2)) = \arg \max_{k_1 \neq k_2 \in [M]} s(k_2, l_{k_1})$. First two patches have the strongest overlapping signal
- For $3 \leq k \leq M$: Let $\hat{\pi}(k) = \arg \max_{j: j \notin L} s(j, \cup_{l \in L} l_l)$. k th patch has the strongest overlapping signal with previous $k - 1$ patches

Theoretical Guarantees

First consider a given patch ordering.

- A1. (Balanced signal across patches) The expectation of each data patch is of rank r and well-conditioned; all patches have similar signal level; cluster sizes are balanced.
- A2. (Sub-Gaussian Noise) The noise matrix

$$E = A_r Z A_c^\top, \quad Z \in \mathbb{R}^{n \times p}, \quad Z_{j,k} \stackrel{\text{ind.}}{\sim} \text{sub-Gaussian}(\sigma) \text{ with zero mean.}$$

Theorem

For any given patch ordering, if regularity conditions (A1, A2) hold, Cluster Quilting achieves consistent clustering: proportion of cluster label mismatch = $o_p(1)$ as long as there is sufficient separation between any cluster centroids, within each feature block:

$$\Delta_m \gg \sigma \|A_r\| \|A_c\| \sqrt{r} K \prod_{l=2}^M \left(\frac{1.1}{\gamma_l} + 1 \right) \left(1 + \sqrt{\frac{p_m}{n_m}} \right),$$

where $\Delta_m = \min_{k \neq k'} \|(\theta_k)_{F_m} - (\theta_{k'})_{F_m}\|_2$, F_m is the feature set observed in m th patch.

First statistical guarantee for clustering with patchwork data!

Minimum separation condition:

$$\Delta_m \gg \sigma \|A_r\| \|A_c\| \sqrt{r} K \prod_{l=2}^M \left(\frac{1.1}{\gamma_l} + 1 \right) \left(1 + \sqrt{\frac{p_m}{n_m}} \right).$$

- With fully observed data and independent noise, this reduces to $\Delta \gg \sigma \sqrt{r} K \left(1 + \sqrt{\frac{p}{n}} \right)$, close to prior work (Löffler et al., 2021) ($\Delta \gg \sigma \sqrt{r} (1 + \frac{p}{n})$).

Minimum separation condition:

$$\Delta_m \gg \sigma \|A_r\| \|A_c\| \sqrt{r} K \prod_{l=2}^M \left(\frac{1.1}{\gamma_l} + 1 \right) \left(1 + \sqrt{\frac{p_m}{n_m}} \right).$$

- Price we pay for patchwork data: factor $\prod_{l=2}^M \left(\frac{1.1}{\gamma_l} + 1 \right)$, where $\gamma_l = \frac{\sigma_r(\text{overlapping btw } l\text{th patch and prior patches})}{\|l\text{th patch}\|}$.
 - $\gamma_l = 0$ if l th patch has no overlapping samples with prior patches;
 - $\gamma_l \geq C$ if l th patch does not introduce new samples.

Theory-guided, Data-driven Patch Ordering

- Theory shows that the ordering affects the clustering performance through $\prod_{m=2}^M \left(\frac{1.1}{\gamma_m} + 1 \right)$
- $\gamma_m = \frac{\sigma_r(\text{overlapping btw } m\text{th patch and prior patches of } X^*)}{m\text{th patch of } X^*}$, an unknown theoretical quantity
- Estimate the merging factor using X ; plug in these estimates into the previously introduced patch-ordering algorithm.
- **The data-driven ordering leads to similar performance to the oracle ordering**, only with an extra factor e^2 in separation condition.

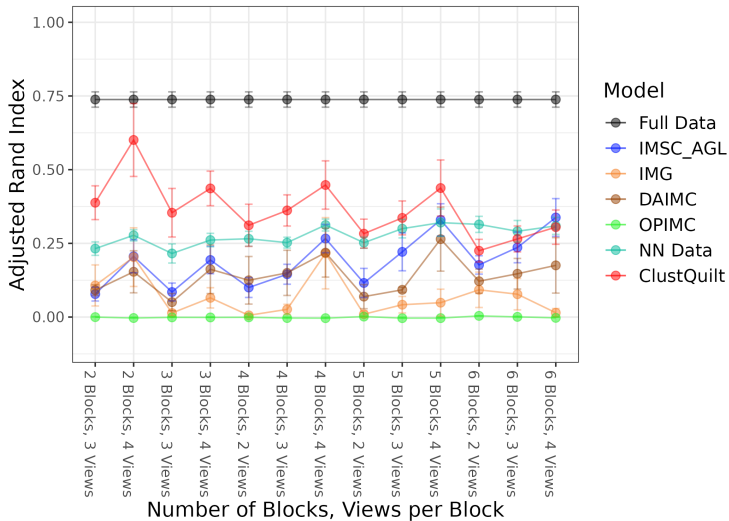
Empirical Studies

- Is our method actually useful in real-world applications?
- Is our method better than other incomplete spectral clustering methods?
- Real-world study from TCGA multiomics data.

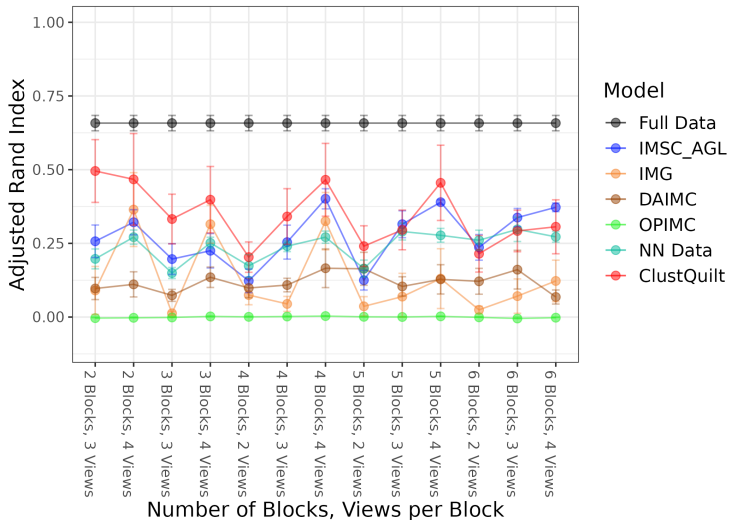
- **The Cancer Genome Atlas (TCGA):** genetic, transcriptomic, and proteomic information for tumors of different cancer types.
- 3 different tumor types (kidney, lung, breast), 2291 observations.
- 5 different views: DNA methylation, miRNA expression, RNAseq gene expression, and reverse phase protein array, 968 features.

- **Create synthetic block missingness from fully observed data.**
 - Randomly partition the rows of the data into distinct observation blocks.
 - Select a subset of views for each block to include in the set of observed entries.
- Vary number of observation blocks , number of views per block.
- 50 replications; mean and SD of adjusted Rand Index.

Multimodal Data Example

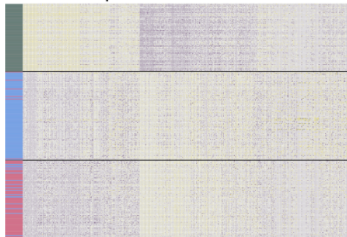


Multimodal Data Example



Multomic Data Example

miRNA Expression



Scaled
Value



miRNA Expression



True
Labels

■ Kidney ■ Breast ■ Lung

CQ estimated clusters; observed data (left) vs. estimated centroids (right).

- Take-away:
 - The patchwork learning paradigm is prevalent but understudied, with various challenges
 - We propose a novel clustering method: cluster quilting, to find **comprehensive clustering structures by stitching the patchwise singular spaces**.
 - **First statistical guarantee** for clustering in this setting
 - [More accurate clustering results across simulations and biomedical data sets!](#)
- Open problems: heterogeneous SNR and cluster sizes across patches? mixed data types?

Acknowledgments

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Andersen Chang



Genevera I. Allen

Thank you!

References

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